

Data Card: Agriculture and Climate Domain-Specific SLM

1. Dataset Overview

This project uses a domain-specific dataset designed to support the development of a Small Language Model (SLM) for agriculture and climate-related question answering.

The dataset combines agricultural and climate knowledge documents with supervised question-answer pairs. Its purpose is to help the model generate coherent, relevant, and context-aware responses to questions related to agricultural practices, climate conditions, crops, and agro-ecological zones.

The dataset is organized into three main components:

- `documents.csv`: agriculture and climate knowledge documents;
- `train_qa.csv`: supervised question-answer examples used for training;
- `test_questions.csv`: unseen questions used for inference and evaluation.

2. Data Sources and Collection

The data used in this project originates from the resources provided for the Agriculture & Climate SLM Challenge.

The knowledge documents contain domain-specific information related to agriculture and climate. The supervised training data links questions and reference answers to relevant agricultural contexts and metadata.

Each training example may include information such as:

- Crop;
- Agro-ecological zone;
- Topic;
- Question;
- Reference answer;
- Document identifier.

The project focuses on structuring these resources into a format suitable for instruction-based fine-tuning.

3. Data Structure

The knowledge component contains documents identified by a unique `document_id`.

The supervised QA component contains examples structured around a question and its corresponding reference answer. Metadata such as crop, agro zone, and topic provides additional context.

During preprocessing, training examples are converted into the following format:

Crop: [crop]
Zone: [agro zone]
Topic: [topic]
Question: [question]
Context: [relevant document text]
Answer: [reference answer]

This structure allows the model to learn the relationship between agricultural context, user questions, and appropriate answers.

4. Data Preprocessing

The preprocessing pipeline includes the following steps:

1. Load the knowledge documents and supervised QA data.
2. Match each training example with its associated document using `document_id`.
3. Extract relevant contextual information from the document.
4. Preserve useful metadata, including crop, agro zone, and topic.
5. Format each example as an instruction-following training sample.
6. Prepare the resulting dataset for Supervised Fine-Tuning (SFT).

Data preprocessing is designed to maintain consistency between the training and inference formats.

5. Intended Use

The dataset is intended for:

- training a domain-specific Small Language Model;
- agricultural and climate question answering;
- experimentation with parameter-efficient fine-tuning methods such as LoRA;
- evaluating approaches for building lightweight AI systems for domain-specific applications.

It is not intended to replace agricultural experts, agronomists, climate scientists, or official advisory services.

6. Data Limitations

The dataset has several limitations.

First, the supervised QA dataset is relatively small, which may limit the model's ability to generalize to questions that differ significantly from the training examples.

Second, agricultural and climate knowledge can vary across locations, seasons, crops, and environmental conditions. A recommendation that is appropriate in one agro-ecological zone may not be appropriate in another.

Third, the model may generate incorrect, incomplete, or overly general responses. Therefore, generated outputs should not be treated as guaranteed expert advice.

7. Data Quality Considerations

Data quality is important because it directly influences model performance.

The project emphasizes:

- consistency between questions and answers;
- relevance of contextual documents;
- preservation of domain metadata;
- verification of document-to-question relationships;
- inspection of formatted training examples.

Future improvements could include additional expert-reviewed agricultural data, broader geographic coverage, and more diverse climate scenarios.